

Notes: Pulvino (JF, 1998)

Do Asset Fire Sales Exist? An Empirical Investigation of Commercial Aircraft Transactions

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1 Big picture

- **Research question.** Do firms forced to liquidate assets because of capital constraints receive prices below fundamental value? More specifically, do financially constrained airlines sell used commercial aircraft at fire-sale discounts?
- **Main idea.** Financial distress can create indirect costs beyond legal and administrative bankruptcy costs. One such cost is that a distressed firm may need cash quickly and may sell assets when the natural buyers are also financially constrained.
- **Theoretical frame.** The paper applies the Shleifer and Vishny liquidation-value logic to commercial aircraft. If assets are industry-specific and the whole industry is depressed, high-value inside buyers may not have enough capital to bid full value. Low-value but well-financed outsiders may then buy at discounts.
- **Empirical setting.** The paper studies used commercial aircraft transactions from 1978 to 1991, focusing on sales and purchases of used narrow-body aircraft by U.S. airlines.
- **Main finding.** Airlines with low spare debt capacity - high leverage and low liquidity - sell aircraft at large discounts relative to hedonic market values. The key discount is about 14 percent in the raw summary and about 13 percent in the regression evidence.
- **State dependence.** Fire-sale discounts are concentrated in market downturns, when the aircraft market is depressed and other airlines are less able to buy.
- **Buyer identity.** Financial institutions and leasing companies pay lower prices than airlines, especially during market downturns. Distressed airlines are also more likely to sell to these outsiders.
- **Buy-side evidence.** Airlines with high spare debt capacity increase purchases when aircraft prices are depressed. Distressed airlines do not. This supports the idea that fire sales create buying opportunities for financially flexible firms.
- **Economic takeaway.** Fire sales are a real indirect cost of financial distress. Maintaining spare debt capacity can be valuable not only because it avoids forced liquidation, but also because it allows firms to buy assets cheaply during industry downturns.

2 Incremental contribution of the paper

- **From announcement effects to transaction prices.** Earlier asset-sale studies often infer liquidation costs from stock-price reactions to sale announcements. Pulvino directly studies the prices at which assets are sold.
- **A cleaner asset class.** Aircraft of the same model, age, and engine stage are relatively homogeneous compared with plants or divisions. This makes transaction-price comparisons more credible.
- **Direct evidence on fire-sale discounts.** The paper estimates discounts using actual sale prices, not only market reactions, accounting signals, or bankruptcy filings.

- **Industry-equilibrium logic.** The paper connects corporate financial distress to market-wide asset liquidity. Liquidation costs are highest when many potential industry buyers are constrained at the same time.
- **Buyer-side validation.** The paper does not only show that constrained sellers get lower prices. It also shows that unconstrained buyers buy more when prices are low.
- **Market thinness.** Commercial aircraft markets are global but still thin: for a given aircraft model, only a small number of transactions occur in a typical month. This gives bargaining power and financial condition room to matter.
- **Capital-structure implications.** The paper provides concrete evidence that leverage can destroy value through asset-sale discounts, strengthening the case for maintaining financial slack in industries with illiquid or industry-specific assets.

3 Data and measurement

3.1 Aircraft transaction data

The main transaction data come from Department of Transportation and Federal Aviation Administration filings assembled by Avmark Inc., an aircraft appraisal and aviation consulting firm.

- **Years.** The main empirical sample covers used aircraft transactions from 1978 to 1991.
- **Asset type.** The core analysis focuses on used narrow-body commercial aircraft.
- **Parties.** Sales and purchases by U.S. airlines are central; transactions by other parties help establish market prices.
- **Recorded transaction details.** The data include aircraft serial number, buyer identity, seller identity, transaction price, transaction date, sale type, engine stage, engine type, and aircraft age.
- **Sample restriction.** The price analysis uses narrow-body models with at least 15 observed transactions between 1978 and 1991.
- **Why stop in 1991?** After the DOT removed its price-disclosure requirement, transaction prices became voluntarily reported and hence vulnerable to selection bias.

Interpretation. The data are unusually well suited to measuring fire-sale discounts because aircraft are real assets with many repeated transactions and observable quality characteristics.

3.2 Airline financial data

Airline financial data come from COMPUSTAT, 10-Ks, 10-Qs, Moody's *Transportation Manual*, and airline operating statistics.

- **Leverage ratio.** Book value of debt plus capitalized lease obligations divided by book debt plus capitalized lease obligations plus book equity.
- **Current ratio.** Current assets divided by current liabilities.

- **Low spare debt capacity, *CAPLO*.** A dummy equal to one if the seller's leverage ratio is above the industry median and its current ratio is below the industry median in the quarter before the sale.
- **High spare debt capacity, *CAPHI*.** A dummy equal to one if the seller's leverage ratio is below the industry median and its current ratio is above the industry median.
- **Debt issues, *ISSUES*.** The number of outstanding debt issues at the end of the fiscal year before the sale. More issues proxy for more dispersed creditors and higher renegotiation costs.

Interpretation. A high-leverage airline is not necessarily forced to sell if it has cash. A low-current-ratio airline is not necessarily deeply constrained if it has little debt. The low-spare-debt-capacity measure combines both long-term debt pressure and short-term liquidity pressure.

3.3 Buyer identity and market conditions

The paper distinguishes buyers by whether they are industry insiders or outsiders.

- **Financial institutions, *FIN*.** A dummy equal to one if the buyer is a financial institution or leasing company.
- **Other buyers, *OTHER*.** A dummy for buyers that are not financial institutions, leasing companies, or major U.S. airlines.
- **Price index, *INDEX*.** A quality-adjusted used-aircraft price index constructed from calendar-quarter dummy coefficients in the hedonic regression.
- **Low *INDEX*.** Periods in the lowest quartile of used-aircraft prices, interpreted as depressed market conditions.
- **High *INDEX*.** Periods in the highest quartile of used-aircraft prices, interpreted as booming market conditions.

Interpretation. Financial institutions can buy aircraft, but they are typically lower-value users than airlines because they must place the aircraft with a lessee before generating operating revenue. Their lower valuation should be most visible when demand for aircraft is weak.

3.4 Firm prospects and aircraft demand controls

The paper includes proxies for seller prospects to separate financial distress from economic distress.

- **Q.** Market value of equity plus book debt divided by book equity plus book debt.
- **REV.** Load factor times revenue per revenue-passenger-mile, measuring the ability to fill planes with high-revenue passengers.
- **COST.** Cost of goods sold divided by available seat miles, measuring cost efficiency.
- **NSALE.** The number of narrow-body aircraft sold by the airline in the same calendar quarter, used to test market thinness.

Interpretation. If low prices simply reflect poor airlines selling low-quality or low-value assets, then firm-prospect controls should absorb the effect. The main fire-sale coefficients remain after these controls.

4 Methodology and empirical specifications

4.1 Shleifer–Vishny motivation

The paper’s organizing mechanism is that the natural buyers of industry-specific assets may themselves be financially constrained during industry downturns.

- If aircraft are best used by airlines, then other airlines are the high-value buyers.
- If the airline industry is depressed, those high-value buyers may face capital constraints.
- A distressed seller may then sell to lower-value but better-financed outsiders.
- The resulting price can be below the asset’s value in its best use.

Prediction. Fire-sale discounts should be strongest when the seller is financially constrained and the used-aircraft market is depressed.

4.2 First-step hedonic pricing model

Pulvino first estimates a hedonic pricing equation for narrow-body aircraft:

$$\log(PRICE) = \beta_0 + \sum_i \beta_i MODEL_i + \sum_j \beta_j QTR_j + \sum_k \beta_k STAGE_k + \beta_{AGE} \log(1 + AGE) + \varepsilon.$$

- $MODEL_i$ controls for aircraft model.
- QTR_j controls for calendar-quarter market conditions.
- $STAGE_k$ controls for engine noise-stage classification.
- AGE controls for the age of the aircraft at the transaction date.

The residual from this regression, RES , is the log difference between the actual transaction price and the quality-adjusted hedonic market price.

Interpretation. A negative residual means the aircraft sold below the price predicted by its model, age, engine stage, and sale quarter.

4.3 Second-step sale-price regression

The main second-step regression is:

$$RES = \beta_0 + \beta_1 CAPLO + \beta_2 CAPHI + \beta_3 ISSUES + \beta_4 (CAPLO \times ISSUES) \\ + \beta_5 FIN + \beta_6 OTHER + \beta_7 Q + \beta_8 REV + \beta_9 COST + \beta_{10} NSALE + \varepsilon.$$

- *CAPLO* tests whether constrained sellers receive lower prices.
- *CAPHI* tests whether financially flexible sellers receive higher prices.
- *CAPLO* $\times$ *ISSUES* tests whether creditor dispersion is especially costly for constrained sellers.
- *FIN* tests whether financial institutions pay lower prices than major U.S. airlines.
- *NSALE* tests whether selling many aircraft in the same quarter depresses prices.

Standard errors. The paper reports both OLS standard errors and conservative Conley-style standard errors that allow correlation among economically close transactions.

4.4 Buyer-identity probit

The paper next asks whether distressed sellers are more likely to sell to financial institutions:

$$FIN = \beta_0 + \beta_1 CAPLO + \beta_2 CAPHI + \beta_3 Q + \beta_4 REV + \beta_5 COST + \beta_6 NSALE + \beta_7 \tau_{industry} + \varepsilon.$$

- The dependent variable equals one if the buyer is a financial institution.
- The key prediction is that low-spare-debt-capacity sellers should be more likely to sell to financial institutions, especially in downturns.
- Industry marginal tax rates are included because financial institutions might value depreciation tax shields differently from airlines.

Interpretation. If distressed sellers are forced to accept bids from lower-value outsiders, financial condition should predict not just price but also buyer type.

4.5 Used-aircraft purchase model

The paper also studies the buy side. The dependent variable is the number of used narrow-body aircraft purchased by airline i in quarter t .

Because purchases are count data with many zeroes and overdispersion, Pulvino uses a negative-binomial maximum likelihood model. The expected number of purchases is specified as:

$$\lambda_{it} = \exp(X_{it-1}\beta).$$

The explanatory variables include financial condition, firm prospects, marginal tax rate, new aircraft deliveries, and fleet size.

- *CAPLO* should predict fewer purchases if financially constrained airlines cannot exploit bargains.
- *CAPHI* should predict more purchases, especially when aircraft prices are depressed.
- New deliveries and owned fleet size control for fleet demand and firm scale.

Interpretation. If distressed sales are true fire sales, financially flexible airlines should show up on the buy side when prices are low.

4.6 Alternative quality tests

A major concern is that distressed airlines may sell lower-quality aircraft, making the apparent discount a quality discount rather than a fire-sale discount.

The paper addresses this concern in several ways:

- It controls for model, age, and engine stage in the hedonic pricing model.
- It reestimates specifications allowing the age premium to vary by quarter.
- It examines a small Braniff subsample with independent appraisals; the appraised-value discount is even larger than the hedonic discount.
- It studies sale/leaseback transactions, where aircraft quality at sale is less likely to affect the up-front price because the airline keeps operating and maintaining the aircraft.
- It tests whether residual variance is larger for older aircraft, where unobserved quality variation should be greater; the test does not support that concern.

Interpretation. Unobserved quality may matter for some transactions, but the paper argues that it cannot explain the full pattern of price discounts, buyer identities, and purchase timing.

5 Identification and interpretation

5.1 What the paper can identify

The paper identifies whether transaction prices, buyer identity, and purchase timing are systematically related to airlines' financial constraints, after controlling for observable aircraft quality and market conditions.

The evidence is strongest as a reduced-form estimate of liquidation costs in a real asset market. It does not recover each aircraft's exact fundamental value, but it shows that constrained sellers receive less for comparable assets, especially when the market is depressed.

5.2 Why aircraft are useful for this question

Commercial aircraft are a useful setting because they are:

- relatively homogeneous within model and age categories;
- frequently traded relative to most industrial plants or divisions;
- valuable enough for discounts to be economically important;
- industry-specific enough that airlines are likely to be higher-value users than outsiders;
- traded in a market that is still thin enough for seller urgency and buyer identity to matter.

Interpretation. If fire-sale discounts appear even in aircraft, which are more liquid than many real assets, then liquidation costs may be even larger for more specialized or less frequently traded assets.

5.3 Financial distress versus economic distress

A key challenge is distinguishing financial distress from weak operating prospects.

- Financial distress is proxied by leverage and liquidity: high leverage plus low current ratio.
- Economic prospects are proxied by Q, revenue-generation variables, and cost variables.
- The main financial-constraint effects survive these controls.

Interpretation. The paper’s evidence is more consistent with financial constraints depressing transaction prices than with low prices merely reflecting bad operating prospects.

5.4 Market downturns as the key state variable

The Shleifer–Vishny logic predicts that fire sales should be most severe when industry insiders are constrained. The paper operationalizes this with the aircraft price index.

- In low-INDEX quarters, aircraft prices are depressed and potential airline buyers are more likely to be financially constrained.
- In high-INDEX quarters, aircraft demand is strong and financially healthy insiders should be more able to bid.
- The main discounts are concentrated in low-INDEX quarters.

Interpretation. Fire sales are not just a property of distressed sellers. They arise from the interaction between distressed sellers and market-wide weak demand or constrained buyers.

5.5 Role of financial institutions

Financial institutions are important in the paper because they represent outside buyers.

- They can provide liquidity when airlines are constrained.
- But they are typically lower-value users because they must find a lessee.
- During downturns, they pay particularly low prices.
- Distressed airlines are especially likely to sell to them in downturns.

Interpretation. The buyer-identity evidence ties the price discount to the mechanism: constrained sellers are pushed toward lower-value but liquid buyers.

5.6 Interpretation of the fire-sale discount

The main coefficient on *CAPLO* should be interpreted as the average discount received by financially constrained sellers relative to otherwise comparable transactions by medium-spare-debt-capacity sellers.

It is not a structural liquidation-value parameter. It is a reduced-form measure of how much less constrained sellers receive in the used-aircraft market, conditional on observed aircraft quality, time, buyer type, and firm controls.

6 Tables and figures

6.1 Table I: Aircraft purchases and sales

Read:

- Table I reports the total number of used narrow-body aircraft purchases and sales by each major U.S. airline in the 1978–1991 sample.
- The table reports 436 purchases and 704 sales in total.
- Eastern Airlines is the largest seller in the table, with 162 sales.
- United, Northwest, Delta, TWA, Braniff, and American also sell large numbers of aircraft.
- The table confirms that there is substantial cross-airline variation in acquisition and liquidation behavior.

Economic message: The setting contains enough repeated aircraft transactions to study how seller financial condition affects sale prices and how buyer financial condition affects acquisition timing.

6.2 Table II: Data summary by spare debt capacity

Read:

- Table II compares transactions by sellers with low, medium, and high spare debt capacity.
- Low-spare-debt-capacity sellers have an average price residual of -0.14, while medium and high spare-debt-capacity sellers have residuals close to zero.
- Low-spare-debt-capacity sellers have higher leverage, lower current ratios, lower coverage ratios, and lower operating income relative to sales.
- Low-spare-debt-capacity sellers also sell younger aircraft, consistent with the need to raise cash.
- Financial institutions buy 54 percent of sales by low-spare-debt-capacity sellers but only 29 percent of sales by high-spare-debt-capacity sellers.

Economic message: The raw data already show the core fire-sale pattern: financially constrained sellers receive lower prices and sell more often to financial institutions.

Table I
Summary of Aircraft Purchase and Sale Transactions
Table entries indicate the total numbers of purchases and sales of used narrow-body aircraft over the 1978–1991 time period.

Airline	Number of Used Narrow-Body Purchases	Number of Used Narrow-Body Sales
Air Florida	11	15
AIRCAL	0	3
Alaska Airlines	17	9
Aloha Airlines	0	3
America West Airlines	16	8
American Airlines	37	48
Braniff Airlines	1	55
Continental Airlines	28	15
Delta Airlines	5	50
Eastern Airlines	47	162
Frontier Airlines	6	0
Hawaiian Airlines	0	4
Midway Airlines	43	6
Muse Air	7	0
New York Air	2	0
Northwest Airlines	10	71
Ozark Airlines	12	2
Pacific Southwest	4	25
Pan Am World Airways	7	33
People Express Airlines	44	0
Piedmont Airlines	28	5
Republic Airlines	21	9
Southwest Airlines	1	0
Trans World Airlines	9	48
United Airlines	38	82
USAIR	36	26
Western Airlines	6	25
Total	436	704

Table I: Summary of aircraft purchases and sales.

Table II
Data Summary

This table summarizes sales of used narrow-body aircraft by selected U.S. airlines from 1978 to 1991. Price residual equals the residual from estimation of equation (1):

$$\log(\text{PRICE}) = \beta_0 + \sum_{i=1}^J \beta_i \text{MODEL}_i + \sum_{j=1}^J \beta_j \text{QTR}_j + \sum_{k=1}^K \beta_k \text{STAGE}_k + \beta_{AGE} \log(1 + \text{AGE}) + \epsilon,$$

where PRICE equals transaction price, MODEL_{*i*} are dummy variables representing aircraft models, QTR_{*j*} are dummy variables representing calendar-quarters, STAGE_{*k*} are dummy variables representing engine stage categories, and AGE equals aircraft age at the time of the transaction. Firms classified as having low spare debt capacities have leverage ratios above the industry median and current ratios below the industry median. Firms classified as having high spare debt capacities have leverage ratios below the industry median and current ratios above the industry median. All other firms are classified as having medium spare debt capacities. Leverage ratio equals total book value of debt divided by the sum of book value of debt and book value of equity; Current ratio equals current assets divided by current liabilities; Coverage ratio equals operating income divided by interest expense where operating income equals income before depreciation, amortization, and taxes; Q equals the sum of market value of equity and book value of debt divided by the sum of book value of equity and book value of debt; Load factor equals the number of revenue-passenger-miles divided by the number of available-seat-miles; COGS equals cost of goods sold. Standard deviations are in parentheses. *t*-statistics for difference-in-means tests are presented between columns.

Variable	Complete Sample (<i>N</i> = 467)	Low Spare Debt Capacity Subsample (<i>N</i> = 144)	<i>t</i> -Statistic (difference between low and medium subsamples)	Medium Spare Debt Capacity Subsample (<i>N</i> = 207)	<i>t</i> -Statistic (difference between medium and high subsamples)	High Spare Debt Capacity Subsample (<i>N</i> = 116)
Price residual	−0.05 (0.41)	−0.14 (0.35)	3.46***	0.001 (0.41)	0.22	−0.01 (0.45)
Leverage ratio (based on book values)	0.68 (0.29)	0.89 (0.30)	8.98***	0.62 (0.24)	5.06***	0.51 (0.15)
Current ratio	0.89 (0.31)	0.70 (0.20)	6.37***	0.86 (0.27)	9.67***	1.17 (0.28)
Coverage ratio	3.65 (4.98)	1.14 (1.52)	8.61***	4.37 (5.08)	1.64	5.49 (6.27)
ln (book value of assets)	7.62 (1.00)	7.23 (1.14)	5.47***	7.84 (0.84)	1.26	7.71 (0.92)
Number of debt issues outstanding	12.09 (6.30)	13.93 (7.24)	2.39*	12.16 (6.21)	4.31***	9.69 (4.07)
Operating income/sales	0.07 (0.06)	0.03 (0.06)	7.94***	0.08 (0.05)	2.47*	0.10 (0.06)
Aircraft age	14.8 (5.26)	13.4 (6.17)	3.45***	15.5 (4.67)	0.55	15.2 (4.72)
Q	1.03 (0.19)	1.08 (0.24)	3.19**	1.01 (0.13)	0	1.01 (0.18)
Load factor × revenue per revenue-passenger-mile	0.08 (0.02)	0.077 (0.02)	3.23**	0.070 (0.02)	5.17***	0.082 (0.02)
COGS/available-seat-mile	0.07 (0.02)	0.068 (0.02)	0.92	0.070 (0.02)	0.43	0.069 (0.02)
Percentage of sales to financial institutions	0.43 (0.50)	0.54 (0.50)	2.03*	0.43 (0.50)	2.54*	0.29 (0.46)

*, **, *** Significant at 5, 1, and 0.1 percent levels, respectively.

Table II: Data summary by seller spare debt capacity.

6.3 Table III: Determinants of sale price

Read:

- Table III is the main sale-price regression table.
- The dependent variable is the residual from the hedonic pricing regression.
- In the complete sample, *CAPLO* is negative and economically large: low-spare-debt-capacity sellers receive about a 13–14 percent discount.
- The interaction between *CAPLO* and debt issues is negative, indicating that creditor dispersion matters most for constrained sellers.
- Sales to financial institutions are priced lower, especially in the low-INDEX subsample.
- In depressed markets, selling to a financial institution is associated with a very large discount; in booming markets the discount largely disappears.
- The number of aircraft sold in a quarter is negatively related to price during depressed markets, consistent with market thinness.

Economic message: The main regression evidence supports fire sales. Financial constraints lower sale prices primarily when the market is weak and natural buyers are likely to be constrained.

Table III
OLS Regressions of the Determinants of Sale Price
This table presents OLS regressions relating narrow-body transaction prices to selling airlines' financial characteristics for 467 aircraft transactions. The dependent variable is the residual from the hedonic regression (equation (1)):

$$\log(\text{PRICE}) = \beta_0 + \sum_{i=1}^4 \beta_i \text{MODEL}_i + \sum_{j=1}^4 \beta_j \text{QTR}_j + \sum_{k=1}^4 \beta_k \text{STAGE}_k + \beta_{\text{CAPLO}} \log(1 + \text{AGE}) + \epsilon,$$

where PRICE equals transaction price, MODEL_{*i*} are dummy variables representing aircraft models, QTR_{*j*} are dummy variables representing calendar-quarters, STAGE_{*k*} are dummy variables representing engine stage categories, and AGE equals aircraft age at the time of the transaction. Panel A presents results using the entire sample, Panel B presents results using the subsample of transactions that occurred when the INDEX of used aircraft prices was in the lowest quartile, and Panel C presents results using the subsample of transactions that occurred when the INDEX of used aircraft prices was in the highest quartile. INDEX is calculated by exponentiating calendar-quarter dummy coefficients; CAPLO is a dummy variable equal to 1 if the selling airline's leverage ratio was above the industry median and its current ratio was below the industry median in the calendar-quarter preceding the transaction; CAPHI is a dummy variable equal to 1 if the selling airline's leverage ratio was below the industry median and its current ratio was above the industry median in the preceding quarter; ISS equals the number of debt issues outstanding in the calendar-quarter preceding the transaction; FIN is a dummy variable equal to 1 if the buyer is a financial or leasing company; OTHER is a dummy variable equal to 1 if the buyer is not a financial institution, leasing company, or large U.S. airline; Q equals market value of equity plus book value of debt divided by book value of equity plus book value of debt; REV equals load factor (equal to revenue-passenger-miles divided by available-seat-miles) times revenue per revenue-passenger-mile; COST equals cost-of-goods-sold divided by available-seat-miles; NSALE equals the number of narrow-body sales by the selling airline in the calendar-quarter of the transaction. Results are relative to transactions where the seller has medium spare debt capacity and the buyer is a major U.S. airline. Both OLS and Conley (1996) standard errors are reported in parentheses below the coefficients.

Table III, part 1: Regression description and variable definitions.

Dependent Variable: Hedonic Regression Residual											
CAPLO	CAPHI	ISS	CAPLO × ISS	FIN	OTHER	Q	REV	COST	NSALE	Const.	Adj R ²
Panel A: Entire Sample (N = 467)											
-0.35 (0.044/0.073)*	-0.04 (0.046/0.088)					0.15 (0.101/0.137)	4.77 (2.57/5.84)	-8.62 (2.63/5.64)	-0.004 (0.006/0.011)	0.09 (0.138/0.170)	0.06
-0.15 (0.044/0.073)*	-0.04 (0.047/0.084)			-0.13 (0.052/0.099)	-0.06 (0.054/0.100)	0.13 (0.102/0.134)	3.68 (2.63/5.92)	7.02 (2.75/5.85)	-0.01 (0.008/0.013)	0.17 (0.133/0.172)	0.07
0.17 (0.090/0.142)	-0.03 (0.047/0.079)	-0.003 (0.004/0.008)	-0.02 (0.006/0.010)*	-0.11 (0.051/0.076)	-0.06 (0.053/0.093)	-0.03 (0.106/0.106)	2.22 (2.62/5.78)	4.23 (2.83/5.35)	-0.002 (0.006/0.010)	0.27 (0.138/0.207)	0.11
Panel B: Low INDEX Quartile (N = 188)											
-0.35 (0.105/0.135)	0.01 (0.011/0.179)					0.59 (0.267/0.305)	3.37 (7.68/8.70)	-11.54 (6.74/7.86)	-0.04 (0.034/0.021)	0.10 (0.438/0.477)	0.11
-0.07 (0.108/0.108)	0.01 (0.098/0.145)			-0.37 (0.113/0.134)**	-0.23 (0.114/0.205)	0.63 (0.302/0.282)	-1.55 (7.63/9.28)	-4.62 (5.90/8.32)	-0.05 (0.055/0.019)*	0.30 (0.453/0.519)	0.16
0.78 (0.267/0.221)**	-0.09 (0.119/0.134)	0.008 (0.008/0.008)	-0.05 (0.015/0.012)**	-0.35 (0.117/0.086)**	-0.27 (0.111/0.179)	-0.02 (0.300/0.283)	11.28 (8.32/9.11)	-6.21 (6.71/7.38)	-0.02 (0.017/0.019)	-0.29 (0.452/0.486)	0.21
Panel C: High INDEX Quartile (N = 104)											
-0.01 (0.092/0.094)	0.08 (0.078/0.164)					-0.17 (0.484/0.545)	-2.42 (5.20/5.90)	1.37 (6.52/8.47)	-0.02 (0.024/0.025)	0.31 (0.459/0.566)	0.00
-0.01 (0.096/0.096)	0.06 (0.084/0.166)			0.04 (0.109/0.096)	0.09 (0.096/0.091)	-0.07 (0.500/0.500)	-0.05 (5.41/5.47)	-0.29 (6.79/7.86)	-0.02 (0.027/0.027)	0.15 (0.484/0.544)	0.01
-0.56 (0.235/0.414)	0.02 (0.074/0.122)	-0.05 (0.010/0.020)*	0.06 (0.018/0.031)	0.14 (0.010/0.113)	0.13 (0.090/0.082)	-0.69 (0.465/0.505)	-3.72 (5.20/5.99)	6.14 (6.24/7.31)	0.004 (0.025/0.026)	0.91 (0.465/0.023)	0.21

*, **, *** Significant at 5, 1, and 0.1 percent levels, respectively.

Table III, part 2: Sale-price regression estimates.

6.4 Table IV: Probability of selling to a financial institution

Read:

- Table IV estimates probit models where the dependent variable equals one if the buyer is a financial institution.
- In the complete sample, low-spare-debt-capacity sellers are more likely to sell to financial institutions, though the full-sample coefficient is imprecise.
- In the low-INDEX subsample, the *CAPLO* coefficient is large and statistically significant.
- The paper reports that moving from high spare debt capacity to low spare debt capacity raises the probability of selling to a financial institution from about 0.14 to 0.49 during depressed markets.
- Firm prospects also matter: economically weaker airlines are more likely to sell to financial institutions during downturns.

Economic message: Distressed sellers do not merely receive lower prices from the same buyer pool. They are more likely to sell to outside buyers, consistent with the market for industry assets becoming illiquid in recessions.

6.5 Figure 1 and Table V: Purchase timing and financial flexibility

Read Figure 1:

- Figure 1 plots purchases, net purchases, and the used-aircraft price index for low- and high-spare-debt-capacity airlines.
- High-spare-debt-capacity airlines increase purchases when used-aircraft prices are depressed.
- The net-purchase panel shows high-spare-debt-capacity airlines as net buyers during depressed markets and net sellers during booms.

- Low-spare-debt-capacity airlines do not display the same countercyclical buying behavior.

Read Table V:

- Table V estimates negative-binomial models for the number of used aircraft purchases per firm-quarter.
- In the complete sample, *CAPLO* is negative and *CAPHI* is positive.
- In low-INDEX quarters, *CAPHI* is large and significant: high-spare-debt-capacity airlines buy far more used aircraft when prices are depressed.
- In high-INDEX quarters, the *CAPHI* effect disappears.
- Marginal tax rates also matter, consistent with the value of depreciation tax shields.

Economic message: The buy-side evidence validates the fire-sale interpretation. Financially flexible airlines behave like buyers of distressed assets when aircraft prices are low.

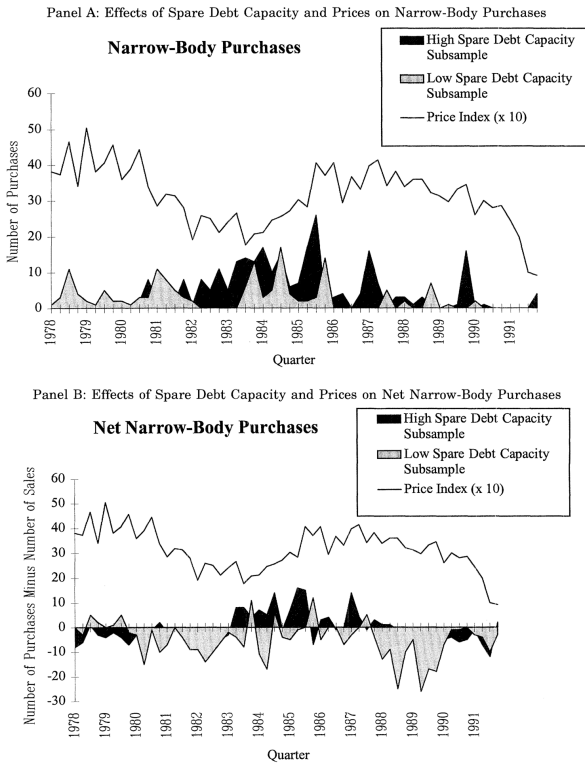


Figure 1. Aircraft acquisition and liquidation summaries for firms with both low and high spare debt capacities. Firm-quarters are segmented based on the quotient of current ratio and leverage ratio. Current ratio equals current assets divided by current liabilities; leverage ratio equals book value of debt divided by book value of debt plus book value of equity. Firm-quarters whose quotient is less than the calendar-quarter's industry median are placed in the low spare debt capacity subsample. Other firm-quarters are placed in the high spare debt capacity subsample. This classification procedure ensures that the two groups have roughly equal numbers of firms.

Figure 1: Acquisition and liquidation patterns by spare debt capacity.

Table V
Maximum Likelihood Estimates for Timing of Used Narrow-Body Purchases
Maximum likelihood estimates presented in this table are based on a Poisson specification with an adjustment to allow for differences between the distribution's mean and variance. The estimated Variance/Mean ratio equals $(1 + \delta)/\delta$. The dependent variable is the number of used narrow-body aircraft purchased in a calendar-quarter. INDEX is calculated by exponentiating calendar-quarter dummy coefficients in the hedonic pricing model described in equation (1):

$$\log(\text{PRICE}) = \beta_0 + \sum_{i=1}^I \beta_i \text{MODEL}_i + \sum_{j=1}^J \beta_j \text{QTR}_j + \sum_{k=1}^K \beta_k \text{STAGE}_k + \beta_{\text{AGE}} \log(1 + \text{AGE}) + \epsilon_t$$

where PRICE equals transaction price, MODEL_i are dummy variables representing aircraft models, QTR_j are dummy variables representing calendar-quarters, STAGE_k are dummy variables representing engine stage categories, and AGE equals aircraft age at the time of the transaction. High values of INDEX correspond to calendar-quarters where prices are relatively high (in the sample's highest quartile). Low values of INDEX correspond to calendar-quarters where prices are low (in the lowest quartile). CAPLO is a dummy variable equal to 1 if the selling airline had a leverage ratio above the industry median and a current ratio below the industry median in the calendar-quarter preceding the transaction; CAPHI is a dummy variable equal to one if the airline had a leverage ratio below the industry median and a current ratio above the industry median in the preceding quarter; Q equals market value of equity plus book value of debt divided by book value of equity plus book value of debt; REV equals load factor (revenue-passenger-miles divided by available-seat-miles) times revenue per revenue-passenger-mile; COST equals cost-of-goods-sold divided by available-seat-miles; τ equals the buyer's marginal tax rate; DELIV equals number of new aircraft deliveries in the current quarter; OWN equals the number of narrow-bodies owned by the airline at the end of the previous quarter. Standard errors are in parentheses.

Model	Dependent Variable = Number of Used Narrow-Body PURCHASES per Firm-Quarter									
	CAPLO	CAPHI	Q	REV	COST	τ	DELIV	log(OWN)	Const.	log(δ)
Complete Sample (<i>N</i> = 1027)	-0.41 (0.238)	0.41 (0.188)*	-0.02 (0.013)	-13.98 (7.95)	13.16 (8.83)	1.27 (0.578)*	-0.07 (0.045)	-0.15 (0.079)	-1.96 (0.516)***	-1.53 (0.167)***
Low INDEX Quartile (<i>N</i> = 293)	0.27 (0.510)	1.07 (0.377)**	0.10 (0.074)	-7.18 (16.04)	-8.22 (17.00)	2.84 (1.04)**	0.01 (0.060)	0.06 (0.192)	-2.96 (1.33)*	-1.71 (0.300)***
High INDEX Quartile (<i>N</i> = 272)	-0.40 (0.417)	0.05 (0.367)	0.01 (0.018)	-19.8 (17.76)	28.0 (20.82)	0.86 (1.21)	-0.09 (0.088)	-0.11 (0.163)	-2.18 (1.06)*	-1.35 (0.313)***

Table V: Timing of used narrow-body purchases.

6.6 Table VI: Sale/leaseback evidence and the quality concern

Read:

- Table VI repeats the price-residual analysis for sale/leaseback transactions.
- Sale/leasebacks are useful because the airline continues to operate and maintain the aircraft, reducing the role of unobserved aircraft quality at the time of sale.
- In the full sample, financial-condition effects are small.
- During 1990–1991, high-spare-debt-capacity airlines receive a significant premium relative to other sellers.
- This supports the idea that financial condition affects transaction terms even when aircraft quality is less likely to explain price differences.

Economic message: The sale/leaseback evidence weakens the alternative explanation that low prices simply reflect lower-quality aircraft sold by distressed airlines.

Table IV

Estimates of the Probability of Selling to a Financial Institution

This table presents probit analyses used to determine whether distressed sellers are more likely to sell to financial institutions. The dependent variable is a dummy variable that takes the value of one if the buyer is a financial institution (i.e., bank or leasing company) and zero otherwise. Panel A presents results using the entire sample, Panel B presents results using the subsample of transactions that occurred when the INDEX of used aircraft prices was in the lowest quartile, and Panel C presents results using the subsample of transactions that occurred when the INDEX of used aircraft prices was in the highest quartile. INDEX is calculated by exponentiating calendar-quarter dummy coefficients in the hedonic pricing model described in equation (1):

$$\log(\text{PRICE}) = \beta_0 + \sum_{i=1}^I \beta_i \text{MODEL}_i + \sum_{j=1}^J \beta_j \text{QTR}_j + \sum_{k=1}^K \beta_k \text{STAGE}_k + \beta_{\text{AGE}} \log(1 + \text{AGE}) + \epsilon,$$

where PRICE equals transaction price, MODEL_i are dummy variables representing aircraft models, QTR_j are dummy variables representing calendar-quarters, STAGE_k are dummy variables representing engine stage categories, and AGE equals aircraft age at the time of the transaction. CAPLO is a dummy variable equal to 1 if the selling airline had a leverage ratio above the industry median and a current ratio below the industry median in the calendar-quarter preceding the transaction; CAPHI is a dummy variable equal to 1 if the airline had a leverage ratio below the industry median and a current ratio above the industry median in the preceding quarter; Q equals market value of equity plus book value of debt divided by book value of equity plus book value of debt; REV equals load factor (revenue-passenger-miles divided by available-seat-miles) times revenue per revenue-passenger-mile; COST equals cost-of-goods-sold divided by available-seat-miles; NSALE equals the number of narrow-body sales by the selling airline in the calendar-quarter of the transaction; τ_{industry} is the weighted average of U.S. airlines' marginal tax rates using firm values as weights. Standard errors are in parentheses and are calculated under the assumption that transactions between airlines are independent. No assumption is made regarding independence of transactions within airlines.

Dependent Variable = 1 if Buyer is a Financial Institution, 0 Otherwise								
CAPLO	CAPHI	Q	REV	COST	NSALE	τ_{industry}	Constant	Pseudo R^2
Panel A: Complete Sample ($N = 467$)								
0.44 (0.335)	-0.34 (0.362)	-1.45 (0.852)	-16.39 (14.26)	31.26* (13.25)	-0.01 (0.067)		0.45 (1.01)	0.08
0.42 (0.335)	-0.32 (0.381)	-1.44 (0.857)	-14.88 (13.99)	28.75 (14.77)	-0.01 (0.067)	-0.62 (1.69)	0.68 (1.13)	0.08
Panel B: Low INDEX Quartile ($N = 168$)								
1.16 (0.429)**	0.11 (0.553)	-0.98 (1.35)	-46.4 (33.40)	54.94 (27.78)*	-0.09 (0.085)		0.66 (2.07)	0.14
1.16 (0.426)**	0.11 (0.555)	-0.97 (1.35)	-47.36 (31.33)	56.53 (27.45)*	-0.10 (0.085)	0.32 (3.05)	0.52 (2.49)	0.14
Panel C: High INDEX Quartile ($N = 104$)								
0.80 (0.510)	0.44 (0.404)	3.23 (1.86)	-25.31 (27.52)	36.55 (34.62)	-0.30 (0.090)***		-3.31 (1.62)*	0.19
0.94 (0.486)	0.87 (0.410)*	5.31 (2.32)*	2.94 (30.84)	-6.65 (42.38)	-0.31 (0.090)***	-5.01 (2.09)*	-3.25 (1.64)*	0.23

*, **, *** Significant at 5, 1, and 0.1 percent levels, respectively.

Table IV: Probability of selling to a financial institution.

Table VI

OLS Regressions of the Determinants of Sale/Leaseback Price

This table presents OLS regressions relating narrow-body sale/leaseback prices to selling airlines' financial characteristics for 524 aircraft transactions. Panel A presents results using the entire sample; Panel B presents results for transactions that occurred during the market downturn in 1990 and 1991. The dependent variable is the residual from the hedonic regression (equation (1)):

$$\log(\text{PRICE}) = \beta_0 + \sum_{i=1}^I \beta_i \text{MODEL}_i + \sum_{j=1}^J \beta_j \text{QTR}_j + \sum_{k=1}^K \beta_k \text{STAGE}_k + \beta_{\text{AGE}} \log(1 + \text{AGE}) + \epsilon,$$

where PRICE equals sale/leaseback price, MODEL_i are dummy variables representing aircraft models, QTR_j are dummy variables representing calendar-quarters, STAGE_k are dummy variables representing engine stage categories, and AGE equals aircraft age at the time of the sale/leaseback. CAPLO is a dummy variable equal to 1 if the selling airline's leverage ratio was above the industry median and its current ratio was below the industry median in the calendar-quarter preceding the transaction; CAPHI is a dummy variable equal to one if the selling airline's leverage ratio was below the industry median and its current ratio was above the industry median in the preceding quarter; Q equals market value of equity plus book value of debt divided by book value of equity plus book value of debt; REV equals load factor (revenue-passenger-miles divided by available-seat-miles) times revenue per revenue-passenger-mile; COST equals cost-of-goods-sold divided by available-seat-miles; NSLB equals the number of sale/leasebacks by the selling airline in the same calendar-quarter of the transaction; τ is the lessee's marginal tax rate. Results are relative to transactions where the seller has medium spare debt capacity. Both OLS and Conley (1996) standard errors are presented in parentheses (OLS standard errors are first). The Conley (1996) procedure corrects for correlation of observations based on proximity in economic distance.

Dependent Variable: Hedonic Regression Residual								
CAPLO	CAPHI	Q	REV	COST	NSLB	τ	Const.	Adj R^2
Panel A: Entire Sample ($N = 524$)								
-0.02 (0.017/0.031)	0.04 (0.019/0.035)	-0.003 (0.046/0.054)	1.61 (1.06/1.73)	-1.31 (1.12/1.65)	-0.001 (0.001/0.001)	-0.10 (0.045/0.061)	-0.002 (0.069/0.091)	0.04
Panel B: 1990–1991 ($N = 71$)								
0.007 (0.082/0.100)	0.18 (0.025/0.059)**	-0.05 (0.086/0.079)	-3.24 (5.77/8.80)	6.63 (5.49/8.06)	-0.002 (0.001/0.002)	0.23 (0.220/0.151)	-0.30 (0.209/0.260)	0.58

Table VI: Sale/leaseback price regressions.

6.7 Appendix Table A.I: Hedonic regression

Read:

- Appendix Table A.I reports the first-step hedonic pricing regression.

- The regression includes aircraft model dummies, quarter dummies, engine-stage dummies, and aircraft age.
- The age coefficient is negative, so older aircraft sell for less.
- The sale regression has an adjusted R^2 of 0.762; the sale/leaseback regression has an adjusted R^2 of 0.947.
- The residuals from this table are the dependent variable in the main sale-price and sale/leaseback price regressions.

Economic message: The hedonic regression provides the quality-adjusted market benchmark used to detect abnormal discounts or premia in transaction prices.

Table AI
Hedonic Regression

This table presents regression results for the model specified in equation (1):

$$\log(\text{PRICE}) = \beta_0 + \sum_{i=1}^I \beta_i \text{MODEL}_i + \sum_{j=1}^J \beta_j \text{QTR}_j + \sum_{k=1}^K \beta_k \text{STAGE}_k + \beta_{\text{AGE}} \log(1 + \text{AGE}) + \epsilon,$$

where PRICE equals transaction price, MODEL_{*i*} are dummy variables representing aircraft models (e.g., DC-9-10), QTR_{*j*} are dummy variables representing calendar-quarters, STAGE_{*k*} are dummy variables representing engine stage categories, and AGE equals aircraft age at the time of the transaction. Dummy variables for Airbus 300B4-200, Stage 3 engines, and the third quarter of 1991 are omitted to avoid collinearity. Column A presents results using sale transactions. Column B presents results using sale/leaseback transactions. Standard errors are in parentheses and are corrected for heteroskedasticity using the methods of Huber (1967) and White (1980). Missing entries are due to lack of transactions and collinearity (e.g., B-737-100 and B-707-320C are the only Stage 1 aircraft in the sample).

Independent Variables	(A) Sales	(B) Sale/Leasebacks	Independent Variables	(A) Sales	(B) Sale/Leasebacks
Model dummies			Quarter dummies		
DC-9-10	-0.40 (0.073)***	-0.03 (0.035)	1978.1	1.34 (0.221)***	—
DC-9-30	0.19 (0.066)**	0.48 (0.039)***	1978.2	1.32 (0.205)***	—
DC-9-50	0.76 (0.092)***	0.69 (0.041)***	1978.3	1.54 (0.247)***	—
B-707-320C	-0.71 (0.151)***	—	1978.4	1.23 (0.199)***	—
B-727-100	-0.63 (0.069)***	-0.03 (0.040)	1979.1	1.62 (0.230)***	—
B-727-100QC	-0.34 (0.071)***	—	1979.2	1.34 (0.214)***	—
B-727-200	-0.07 (0.081)	0.32 (0.028)***	1979.3	1.40 (0.219)***	—
B-727-200 ADV	0.46 (0.077)***	0.64 (0.049)***	1979.4	1.52 (0.211)***	—
B-737-100	—	0.47 (0.050)***	1980.1	1.28 (0.200)***	—
B-737-200	0.14 (0.073)*	0.46 (0.036)***	1980.2	1.36 (0.198)***	—
B-737-200 ADV	0.77 (0.110)***	0.79 (0.049)***	1980.3	1.49 (0.230)***	—
B-737-300	-0.51 (0.170)***	-0.53 (0.038)***	1980.4	1.22 (0.192)***	—

Table AI—Continued

Independent Variables	(A) Sales	(B) Sale/Leasebacks	Independent Variables	(A) Sales	(B) Sale/Leasebacks
Quarter dummies			Quarter Dummies		
1981.1	1.05 (0.192)***	—	1987.2	1.42 (0.206)***	0.13 (0.037)***
1981.2	1.16 (0.197)***	—	1987.3	1.23 (0.186)***	0.18 (0.036)***
1981.3	1.1 (0.202)***	—	1987.4	1.34 (0.190)***	0.17 (0.038)**
1981.4	1.03 (0.200)***	0.04 (0.038)	1988.1	1.22 (0.236)***	-0.59 (0.217)**
1982.1	0.65 (0.226)**	—	1988.2	1.28 (0.200)***	0.06 (0.044)
1982.2	0.95 (0.208)***	—	1988.3	1.28 (0.191)***	0.11 (0.037)**
1982.3	0.92 (0.223)***	—	1988.4	1.17 (0.196)***	0.23 (0.041)***
1982.4	0.75 (0.240)***	0.09 (0.037)*	1989.1	1.14 (0.195)***	0.24 (0.043)***
1983.1	0.88 (0.226)***	—	1989.2	1.09 (0.189)***	-0.31 (0.038)***
1983.2	0.98 (0.202)***	—	1989.3	1.20 (0.197)***	0.21 (0.034)***
1983.3	0.57 (0.220)**	0.10 (0.037)**	1989.4	1.24 (0.185)***	0.17 (0.041)***
1983.4	0.73 (0.194)***	-0.16 (0.050)***	1990.1	0.96 (0.222)***	0.15 (0.041)***
1984.1	0.75 (0.191)***	-0.28 (0.043)***	1990.2	1.10 (0.192)***	0.23 (0.034)***
1984.2	0.90 (0.193)***	-0.09 (0.044)	1990.3	1.03 (0.203)***	-0.09 (0.045)*
1984.3	0.94 (0.211)***	-0.01 (0.038)	1990.4	1.05 (0.214)***	0.09 (0.022)***
1984.4	1.00 (0.192)***	0.19 (0.048)***	1991.1	0.90 (0.195)***	0.20 (0.039)***
1985.1	1.11 (0.190)***	0.11 (0.036)**	1991.2	0.69 (0.214)**	-0.002 (0.001)
1985.2	1.04 (0.192)***	0.16 (0.12)	1991.4	-0.08 (0.409)	0.38 (0.034)***
1985.3	1.40 (0.193)***	0.07 (0.023)**	log(1 + AGE)	-0.16 (0.057)***	-0.29 (0.020)***
1985.4	1.31 (0.193)***	0.29 (0.040)***	St1 Dummy	-2.16 (0.277)***	—
1986.1	1.40 (0.206)***	0.22 (0.033)***	St2 Dummy	-1.73 (0.146)***	-1.56 (0.048)***
1986.2	1.08 (0.243)***	0.04 (0.056)	Constant	2.83 (0.247)***	3.82 (0.056)***
1986.3	1.30 (0.221)***	0.15 (0.041)***			
1986.4	1.20 (0.195)***	0.18 (0.040)***	Number of observations	1079	621
1987.1	1.38 (0.191)***	0.10 (0.038)**	Adjusted R^2	0.762	0.947

*, **, *** Significant at the 5, 1, and 0.1 percent levels, respectively.

Appendix Table A.I, part 1: Hedonic regression, model and early quarter dummies.

Appendix Table A.I, part 2: Continued hedonic regression and fit statistics.

6.8 Appendix Table B.I: Conley standard errors

Read:

- Appendix Table B.I describes the weighting scheme used to compute Conley-style standard errors.

- Transactions by the same seller, same buyer, same model, and same date receive the highest correlation weight.
- Transactions by the same seller receive positive weights even if they are separated in time.
- These conservative standard errors are usually larger than OLS standard errors.

Economic message: The paper recognizes that aircraft transactions are not independent draws. A seller liquidating a fleet may have correlated residuals across multiple transactions, so standard errors should allow for economic proximity among sales.

Table BI
Conley Standard Errors

Weighting Factor	Condition
$w_{ij} = 1.0$	$i = j$
$w_{ij} = 1.0$	$\text{Seller}_i = \text{Seller}_j \ \& \ \text{Buyer}_i = \text{Buyer}_j \ \& \ \text{Model}_i = \text{Model}_j \ \& \ \text{Date}_i = \text{Date}_j$
$w_{ij} = 0.75$	$\text{Seller}_i = \text{Seller}_j \ \& \ \text{Date}_i - \text{Date}_j < 1 \text{ year}$
$w_{ij} = 0.50$	$\text{Seller}_i = \text{Seller}_j$
$w_{ij} = 0.0$	Otherwise

Appendix Table B.I: Weighting scheme for Conley standard errors.

7 Short summary

- The paper asks whether asset fire sales exist by studying commercial aircraft transactions.
- It uses transaction prices for used narrow-body aircraft from 1978 to 1991.
- A hedonic pricing model controls for aircraft model, age, engine stage, and calendar-quarter market conditions.
- The main outcome is the residual from this hedonic pricing model.
- Airlines with high leverage and low liquidity sell aircraft at substantial discounts.
- These discounts are concentrated in depressed used-aircraft markets.
- Financial institutions and leasing companies pay lower prices than airlines during downturns.
- Distressed airlines are more likely to sell to financial institutions when the market is depressed.
- Airlines with high spare debt capacity buy more used aircraft when aircraft prices are low.
- Sale/leaseback evidence and additional robustness checks suggest that unobserved aircraft quality is unlikely to explain the results.
- The paper provides direct evidence that fire-sale liquidation discounts are an indirect cost of financial distress.

8 Conclusion

8.1 Core conclusion

Pulvino shows that asset fire sales exist in the commercial aircraft market. Financially constrained airlines sell used narrow-body aircraft at lower prices than more conservatively financed airlines, especially when aircraft markets are depressed. These discounts are economically large and consistent with the idea that distressed sellers accept prices below fundamental value when high-value industry buyers are also constrained.

8.2 Economic intuition

The highest-value buyer for an aircraft is usually another airline. But if the whole airline industry is in distress, many airlines cannot finance new purchases. A distressed seller that needs cash quickly may then have to sell to a financial institution or leasing company. These buyers can provide liquidity, but they are lower-value users because they must find a lessee before the aircraft generates revenue. The seller therefore receives a lower price.

8.3 Incremental contribution revisited

The paper’s key contribution is to estimate liquidation discounts from actual real-asset transaction prices. It moves beyond announcement effects and directly links seller financial condition, buyer identity, and acquisition timing to asset-sale prices. This combination makes the evidence stronger than a price-discount regression alone.

8.4 Implications for capital structure

The findings imply that debt has indirect costs. A highly levered firm may not only face legal and administrative costs in distress; it may also be forced to sell assets at discounts. Financial slack is valuable because it helps a firm avoid being a distressed seller and allows it to become a buyer when competitors are forced to sell.

8.5 Implications for investment and bankruptcy policy

The evidence also implies that asset liquidation can be costly. If liquidation transfers assets to lower-value users, immediate cash liquidation may fail to maximize creditor recoveries and may allocate assets inefficiently. This matters for debates over whether distressed firms should be reorganized or liquidated quickly.

8.6 Limitations

The paper cannot observe each aircraft’s true fundamental value. Some unobserved quality differences may remain despite controls for model, age, and engine stage. The identification is observational, based on financial condition and market states rather than randomized distress. However, the consistency of the sale-price, buyer-identity, buy-side, and sale/leaseback evidence makes the fire-sale interpretation persuasive.

8.7 Takeaway

The enduring takeaway is that financial distress can force firms to sell real assets at discounts. Fire-sale costs are largest when industry conditions are weak and potential high-value buyers are financially constrained. Maintaining spare debt capacity is valuable because it protects firms from selling low and positions them to buy when others are forced to liquidate.